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Toward Psychometric Learning Analytics: Augmenting the Urnings Algorithm with Response Times

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Adaptive learning systems (ALS) aim to tailor the educational material to match the student's needs, ultimately improving the learning outcomes. An ALS dynamically adjust the level of practice based on the student's ability; therefore, obtaining accurate ability estimates is crucial. Since the amount of responses in a timeframe is limited, high measurement precision is unattainable using only accuracies, which calls for the inclusion of other data sources into the measurement. Here, we propose algorithms that can estimate the abilities on-the-fly based on both accuracy and response times (RT). These are extensions of a rating system called the Urnings algorithm. Since the Urnings algorithm uses discrete updates, building on the difference between a single observed and simulated response, we combined accuracy and RT into a continuous score using a discretized version of the Signed Residual Time (SRT) scoring rule. Through simulation studies, we showed that by augmenting the algorithm with RT, a reliable ability measure and better ability tracking can be obtained by administering fewer items. By reanalyzing data from an existing ALS, we showed that the algorithms can be utilized even if the SRT scoring rule is not explicitly used during measurement, providing better ability estimates and smaller prediction errors.

Introduction

One of the most promising ambitions in educational technology is the move toward large-scale personalized learning enabled by the development of adaptive learning systems (ALS) (Brusilovsky et al., 1995; Vermeiren et al., 2024). An ALS dynamically adjust the level of practice based on the ability of the student at a given time, thus personalizing the learning content and providing an optimally engaging environment for the students (Ennouamani & Mahani, 2017; Kabudi et al., 2021). To enable this, one needs a continuously updated, accurate and reliable measure of the possibly changing student ability. However, an ALS's adaptive, large-scale, anytime-anywhere nature poses challenges to traditional estimation techniques (Wauters et al., 2010; Yan et al., 2021). Recently computationally inexpensive methods were developed to account for the large streams of data an ALS produces, building on a sequential updating scheme based on the observed and expected response on practice items, that is, rating systems such as the Elo rating system (ERS) (Bolsinova et al.,

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2024; Klinkenberg et al., 2011; Pelánek, 2016) and the Urnings algorithm (Bolsinova et al., 2022; Brinkhuis & Maris, 2019). Rating systems provide a computationally efficient alternative to estimating dynamic measurement models (Embretson, 1991; Wang et al., 2013), and they are capable of tracking the ability development without assuming a specific growth model, which is advantageous due to the lack of formal theories about how learning progresses (Vermeiren et al., 2025). They also enable the on-the-fly estimation of item difficulties as opposed to Computerized Adaptive Testing, which is necessary since the pre-calibration of the possibly vast number of items is impractical (van der Linden et al., 2000).

Still, an important challenge remains: although ALS generate a large stream of data, zooming in on a particular person at a particular time, the amount of data is limited, which makes high measurement precision currently unattainable using response accuracy data (Bolsinova et al., 2024; Gergely et al., 2023, 2024; Vermeiren et al., 2024).

Measurement precision is a crucial issue to tackle for multiple reasons. First, improving precision is needed to achieve sufficient power to detect ability change at the individual level (Dumas & McNeish, 2017), which is important to evaluate the effectiveness of educational interventions and to test hypotheses about how idiographic ability development unfolds (Ding, 2009; Koopmans, 2014; Savi et al., 2021). Second, if measurement within ALS is of sufficient quality, then learning and assessment can be integrated, significantly reducing the length of testing (Deonovic et al., 2018; von Davier, 2017). Third, less uncertainty allows for better adaptation of the learning material to the student's level, such that they are neither over-challenged nor under-challenged. In cases like this, the motivation of the students can decrease, which could lead to dropping out of the ALS (Akhtar et al., 2023; Akhtar & Kovacs, 2024; Brinkhuis et al., 2018; Jansen et al., 2016).

In standard psychometric settings, there are two ways of increasing precision: administering more items (Brown, 1910) and/or optimizing item information (Weiss, 1982). In ALS, the system does not have full control over how often and how long students practice. Selecting items with maximum information, as is done in computerized adaptive testing (van der Linden et al., 2000; van der Linden, 2009), results in presenting items with a relatively low probability of a correct response, which has negative consequences for student motivation (Akhtar et al., 2023). In the case of an ALS, studies recommended selecting items with a success rate well above 50% (Brinkhuis et al., 2018; Jansen et al., 2016).

Due to the computerized nature of ALS, additional relevant information is available beyond the accuracy of item responses, which are usually referred to as process data (e.g., response times) (van der Linden, 2008), eye and mouse tracking data (Barrios et al., 2004; de Mooij et al., 2021), hint usage (Bolsinova et al., 2022). Response time (RT) has been readily found to be an important source of information about student performance (Luce, 1991); however, modeling RT and accuracy has been highly challenging (Maris & van der Maas, 2012; van der Linden, 2009). van der Linden (2007) introduced the hierarchical framework for response time and accuracy. One of the goals of this approach is to improve the precision of ability estimates by borrowing information from the measurement of speed (a latent variable related to RTs) for the measurement of ability when these two latent variables

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are correlated. This correlation is estimated from data, thus, how RT influences ability estimation is not clear before data collection. In practical applications, positive (Wang & Xu, 2015), negative (Scherer et al., 2015), and close to zero (Shaw et al., 2020) correlations were observed. The data-driven nature of the hierarchical framework is undesirable from the perspective of fairness and transparency since students with problem-solving strategies different from the majority of students will be disadvantaged (Tijmstra & Bolsinova, 2022).

An alternative approach proposed by Maris and van der Maas (2012) builds on the idea of transparency. RTs are incorporated into the measurement of ability by making them part of an explicit scoring rule called Signed Residual Time (SRT). Compared to the Rasch model's scoring rule, where 1/0 (or equivalently 1/–1) scores are assigned to correct/incorrect responses, the signed-residual time (SRT) scoring rule makes the positive contribution of a response RT-dependent (i.e., fast correct responses are taken to represent higher ability), while also penalizes fast incorrect responses. Similar to how the Rasch model follows from the assumption that the student's sum score is a sufficient statistic for ability (Fischer, 1995), a measurement model can be formulated from the SRT scoring rule. This approach makes ability estimation transparent and easy to explain to stakeholders (students, parents, teachers) as opposed to using data-driven approaches that optimize prediction but lack interpretability. It also allows for keeping the model relatively simple, which is important to make on-the-fly estimation of ability feasible.

The SRT scoring rule has been applied to the Elo rating system in large-scale ALS, such as MathGarden (Brinkhuis et al., 2018; Klinkenberg et al., 2011), by building the updating scheme on the observed and expected SRT scores. However, ERS has important shortcomings. The estimates of ability in Elo have no known limiting distribution and consequently their standard errors are not available, making individual-level inference difficult. Even in simple cases, where items are selected randomly, the estimates are biased (Bolsinova et al., 2024), while when adaptive item selection is used based on previous estimates, the estimates in the system diverge from each other (Bolsinova et al., 2024; Hofman et al., 2020). The Urnings algorithm, which was developed to overcome these limitations (Bolsinova et al., 2022), has superior measurement properties and is therefore used in the current study. The Urnings algorithm uses discrete updates based on the difference between the observed and a simulated response accuracy, where the simulation is based on the expected value. This implies that the Urnings algorithm is, at its very core, a discrete state Markov chain, thus, immediate implementation of SRT is not possible. Deonovic et al. (2020) outlined a discretization strategy to include continuous variables into an Urnings-type updating scheme. However, it is not trivial how their algorithm could be optimally applied to ALS.

The goal of the current study is to develop RT-based extensions to the Urnings algorithm suitable for applications in the ALS. Building on the discretization strategy outlined by Deonovic et al. (2020), we develop two different updating schemes. These new algorithms simplify the proposed rating system of (Deonovic et al., 2020), enabling adaptive item selection based on the ratings, and providing a way to constrain the item difficulties such that the change of student abilities is identifiable. Our general research question is how the quality of measurement in

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the ALS changes when our extended algorithm is used, compared to using response accuracy data only. More specifically, in a simulation study, we evaluate whether the estimates of ability converge faster and whether they are better able to follow change when RTs are used, and in an empirical example, we evaluate whether the accuracy of prediction of the next response is better when the extended algorithms are used compared to the basic Urnings algorithm.

Methods

Urnings Algorithm

The Urnings algorithm (Bolsinova et al., 2022) is a way to dynamically estimate the parameters of a Rasch model by constructing an ergodic Markov chain with finite and discrete state space, thus the ability and difficulty estimates have a unique limiting distribution, are unbiased and have known standard errors.

First, let X_{pi} be a binary response variable (1—correct, 0—incorrect), θ_p the ability of student p and δ_i, the difficulty of item i. To move to a discrete state space for parameter estimates, we first transform θ_p and δ_i onto the [0,1] interval by defining expit-transformed parameters as $\pi_p = \frac{\exp(\theta_p)}{1+\exp(\theta_p)}$ and $\xi_i = \frac{\exp(\delta_i)}{1+\exp(\delta_i)}$, thus the Rasch model can be written as:

$$P(X_{pi} = 1 \mid \pi_p, \xi_i) = \frac{\pi_p(1-\xi_i)}{\pi_p(1-\xi_i) + \xi_i(1-\pi_p)}. \tag{1}$$

The probability above is analogous to an urn game (Johnson & Kotz, 1991). Both a student's ability and an item's difficulty can be characterized by infinitely large urns containing green and red balls, where π_p and ξ_i are the proportion of green balls in the corresponding urns. The probability of the student correctly solving the item is equal to drawing a green ball from the ability urn and a red ball from the difficulty urn, where draws of the same color are discarded.

Instead of creating an updating scheme for the continuous variables π_p and ξ_i we define two discrete variables which can take values from a grid of proportions as $\{\frac{0}{n}, \frac{1}{n}, \ldots, \frac{n-1}{n}, \frac{n}{n}\}$, where n is the granularity of the grid. These variables can be viewed as urns with a finite amount of red and green balls. The urn sizes can differ among students and items and will be denoted as n_p and n_i, respectively. We denote the number of green balls in the student's urn as $R_p \in \{0, 1, \ldots, n_p\}$ and the item's urn as $U_i \in \{0, 1, \ldots, n_i\}$. We refer to the proportions $\frac{R_p}{n_p}$ and $\frac{U_i}{n_i}$ as ratings.

Now the ratings $\frac{R_{p,t}}{n_p}$ and $\frac{U_{i,t}}{n_i}$ at timepoint t can be obtained as

$$\begin{aligned} \frac{R_{p,t}}{n_p} &= \frac{R_{p,t-1}}{n_p} + \frac{1}{n_p}(X_{pi} - X^*_{pi}), \\ \frac{U_{i,t}}{n_i} &= \frac{U_{i,t-1}}{n_i} + \frac{1}{n_i}(X^*_{pi} - X_{pi}), \end{aligned} \tag{2}$$

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where X^*_{pi} is a simulated response obtained by drawing a Bernoulli random variable with a probability calculated using the ratings at time $t-1$ augmented by the observed response X_{pi} as

$$X^*_{pi} \sim Bernoulli\left(\frac{\frac{R_{p,t-1}+X_{pi}}{n_p+1}\left(1-\frac{U_{i,t-1}+(1-X_{pi})}{n_i+1}\right)}{\frac{R_{p,t-1}+X_{pi}}{n_p+1}\left(1-\frac{U_{i,t-1}+(1-X_{pi})}{n_i+1}\right)+\frac{U_{i,t-1}+(1-X_{pi})}{n_i+1}\left(1-\frac{R_{p,t-1}+X_{pi}}{n_p+1}\right)}\right). \quad (3)$$

The probability used to simulate X^*_{pi} can be seen as the result of the following urn game: First, we add a green ball to the student's urn and a red ball to the item's urn if the response was correct, and vice versa if the response was incorrect. Second, we draw one ball from the student's urn and one ball from the item's urn until they are of a different color. The probability of drawing $X^*_{pi}=1$ is equal to drawing a green ball from the student's augmented urn and a red ball from the item's augmented urn, where draws of the same color are discarded. The updating scheme defined in Equation 2 can update the student ratings by $\frac{1}{n_p}$, 0 or $-\frac{1}{n_p}$ with corresponding symmetric item updates by $-\frac{1}{n_i}$, 0 and $\frac{1}{n_i}$.

Based on ratings, it is possible to adaptively select items for the student. In other rating systems, using adaptive item selection increases the variance of the student ratings over time, thus, no limiting distribution can be found. The Urnings algorithm solves this issue by introducing a Metropolis-Hastings algorithm, which rejects some of the proposed updates. The probability of accepting a proposed update is

$$\alpha = \min\left(1, \frac{p(i \mid R_{p,t}, n_p, R_{i,t}, n_i)}{p(i \mid R_{p,t-1}, n_p, R_{i,t-1}, n_i)}\right), \quad (4)$$

where $p(i \mid R_{p,t-1}, n_p, R_{i,t-1}, n_i)$ is the probability of selecting item i for the student p based on their current rating and $p(i \mid R_{p,t}, n_p, R_{i,t}, n_i)$ is selection probability after the proposed update was done.

Using the outlined algorithm, the ratings converge to a product of non-identical binomial distributions, conditional on the total sum of the ratings in the system, with the urn size being the number of trials and the ability and difficulty parameters, the probabilities such as:

$$p(\mathbf{R}, \mathbf{U}) = \frac{\prod_{k=1}^{K}\binom{n_k}{R_k}\pi_k^{R_k}(1-\pi_k)^{n_k-R_k}\prod_{l=1}^{L}\binom{m_l}{U_l}\xi_l^{U_l}(1-\xi_l)^{m_l-U_l}}{Z}\mathbf{I}_{\sum_K R_k+\sum_L U_l=C}, \quad (5)$$

where K and L are the number of students and items respectively, $\mathbf{R}$ and $\mathbf{U}$ are vectors of student and item ratings, $\mathbf{I}_{\sum_K R_k+\sum_L U_l=C}$ is an indicator function valued 1 if the sum of all ratings is equal to a constant C and 0 otherwise and Z is the normalizing constant (Bolsinova et al., 2022; Gergely et al., 2024). For large systems, the limiting distribution approaches a product of non-identical binomial distributions (find the proof in Bolsinova et al. (2022) (p. 20-23). The estimates are unbiased as they are the expected values of the corresponding binomial distributions. The variance and standard errors are readily calculated, e.g. for person i the variance is $n_i\frac{R_{i,t}}{n_i}\left(1-\frac{R_{i,t}}{n_i}\right)$ and the standard error is $\left[\frac{R_{i,t}}{n_i}\left(1-\frac{R_{i,t}}{n_i}\right)\right]^{1/2} n_i^{-1/2}$ (Brown et al., 2001; Gergely et al., 2024).

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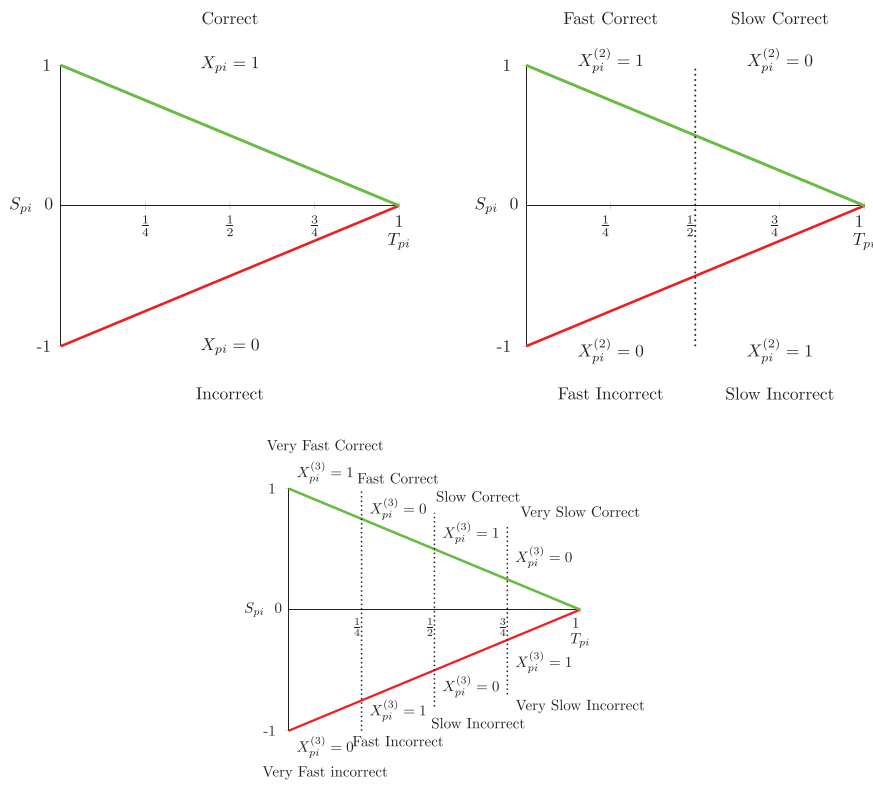


Figure 1. Visual representation of 3 levels of dyadic expansions. Plot (a) describes how the accuracy is decoupled from the ratings, while plots (b) and (c) present how the first and second pseudo-RT are defined.

Signed Residual Time Model

Now, the Urnings algorithm builds its updating scheme on the difference between a single observed and simulated outcome, so if one wants to incorporate RT into the measurement, it comes naturally to use explicit scoring rules as they combine accuracy and response time in a single variable. Here we consider the SRT scoring rule. Let T_{pi} be the proportion of time elapsed from the available time limit until student p responded to item i. The SRT scoring rule can be formulated as:

$$S_{pi} = (2X_{pi} - 1)(1 - T_{pi}), \tag{6}$$

where S_{pi} is the score, which can take values between -1 and 1 (see Figure 1a).

Maris and van der Maas (2012) showed that building on SRT it is possible to construct an IRT model, which belongs to the exponential family assuming that the total SRT score of a student is a sufficient statistic for their ability (X_p, $T_p \perp\!\!\!\perp \theta_p \mid \sum_i S_{pi}$) and the total SRT score of an item is a sufficient statistic for its difficulty (X_i, $T_i \perp\!\!\!\perp$

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$\delta_i \mid \sum_p S_{pi}$). Taking these assumptions together, the joint probability density function of accuracy and RT for observing is:

$$p(X_{pi}, T_{pi} \mid \theta_p, \delta_i) = \frac{\exp\left((2X_{pi} - 1)(1 - T_{pi})(\theta_p - \delta_i)\right)}{\sum_{j=0}^{1} \int_0^1 \exp\left((2j - 1)(1 - l)(\theta_p - \delta_i)\right) dl}. \tag{7}$$

Discretization of the Continuous Scores

Since the SRT score is a continuous variable and the Urnings algorithm was developed for discrete responses, we must create a discretized approximation of the continuous measure. Deonovic et al. (2020) discretized the SRT scores using dyadic expansions. The main idea behind dyadic expansions is to use a limited number of conditionally independent binary indicator variables derived from the score (Billingsley, 2017).

To motivate this discretization, we can think about how students can receive the most points in different situations under the SRT scoring rule. The SRT rule assigns a higher score for students who solved a given item correctly compared to those who solved it incorrectly (see Figure 1a). If the response is correct, a faster student gets more points compared to a slower one, whereas if the response is incorrect, but the student spent more time trying to solve the item, their score will be higher than a fast and incorrect response, to discourage guessing (see Figure 1b). This way, it comes naturally to first decouple accuracy from the score and then define a pseudo-RT indicator variable independent of the accuracy. To follow the logic of the original model, this pseudo-RT variable needs to be defined such that it values fast correct and slow incorrect responses higher compared to slow correct and fast incorrect respectively. Now we turn to formally defining these indicator variables.

First, we need to calculate the marginal probability distribution for accuracy as

$$\begin{aligned} p(X_{pi} = 1 \mid \theta_p, \delta_i) &= \int_0^1 \frac{\exp\left((1 - t)(\theta_p - \delta_i)\right)}{\sum_{j=0}^{1} \int_0^1 \exp\left((2j - 1)(1 - l)(\theta_p - \delta_i)\right) dl} dt \\ &= \frac{\exp\left(\theta_p - \delta_i\right)}{1 + \exp\left(\theta_p - \delta_i\right)}, \end{aligned} \tag{8}$$

which is a Rasch model (or one-parameter logistic model (1PL) with item discrimination equal to 1).[1] To introduce the pseudo-RT variable, first we define two categories of responses based on their RTs: fast ($T_{pi} < \frac{1}{2}$) and slow ($T_{pi} > \frac{1}{2}$). Let's look at the probability of observing a fast response given that the response is correct and the probability of observing a slow response given that the response is incorrect. After simplifying with the normalizing constant of the joint probability distribution, we obtain

$$\begin{aligned} p\left(T_{pi} < \frac{1}{2} \mid X_{pi} = 1\right) &= \frac{\int_0^{\frac{1}{2}} \exp\left((1 - t)(\theta_p - \delta_i)\right) dt}{\int_0^1 \exp\left((1 - t)(\theta_p - \delta_i)\right) dt}, \\ p\left(T_{pi} > \frac{1}{2} \mid X_{pi} = 0\right) &= \frac{\int_{\frac{1}{2}}^1 \exp\left((t - 1)(\theta_p - \delta_i)\right) dt}{\int_0^1 \exp\left((t - 1)(\theta_p - \delta_i)\right) dt}. \end{aligned} \tag{9}$$

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If we solve the integrals and simplify the formulas,[2] we can derive that the two probabilities are equal to each other:

$$p\left(T_{pi} < \frac{1}{2} \mid X_{pi} = 1\right) = p\left(T_{pi} > \frac{1}{2} \mid X_{pi} = 0\right) = \frac{\exp\left(\frac{1}{2}(\theta_p - \delta_i)\right)}{1 + \exp\left(\frac{1}{2}(\theta_p - \delta_i)\right)}. \quad (10)$$

Similarly, it can be shown that the probability of observing a slow correct and a fast incorrect response is equal to one minus the probability in Equation 10 thus, we can define the pseudo-RT indicator variable as:

$$X_{pi}^{(2)} = \begin{cases} 1 & \text{if } T_{pi} < \frac{1}{2}, X_{pi} = 1 \text{ or } T_{pi} > \frac{1}{2}, X_{pi} = 0 \\ 0 & \text{if } T_{pi} > \frac{1}{2}, X_{pi} = 1 \text{ or } T_{pi} < \frac{1}{2}, X_{pi} = 0. \end{cases} \quad (11)$$

X_{pi} and $X_{pi}^{(2)}$ are conditionally independent given θ_p and δ_i, hence they provide independent sources of information about ability, and $X_{pi}^{(2)}$ follows the same model as X_{pi} but with discrimination of $\frac{1}{2}$.

We can continue this dyadic expansion by further dividing the responses into categories as: very fast, fast, slow and very slow (see Figure 1c). Here, it can be shown that

$$P\left(T_{pi} < \frac{1}{4} \mid X_{pi} = 1, T_{pi} < 1/2\right) = P\left(\frac{1}{2} < T_{pi} < \frac{3}{4} \mid X_{pi} = 1, T_{pi} > 1/2\right) =$$
$$P\left(T_{pi} > \frac{3}{4} \mid X_{pi} = 0, T_{pi} > 1/2\right) = P\left(\frac{1}{4} < T_{pi} < \frac{1}{2} \mid X_{pi} = 0, T_{pi} < 1/2\right). \quad (12)$$

Using this, we can define $X_{pi}^{(3)}$, which takes the value 1 for very-fast-correct, slow-correct, very slow-incorrect, and fast-incorrect responses and 0 otherwise. $X_{pi}^{(3)}$ is conditionally independent of X_{pi} and $X_{pi}^{(2)}$ and follows the same 1PL model, but with a discrimination of $\frac{1}{4}$. This process could continue indefinitely introducing an array of dichotomous variables all conditionally independent given θ_p and δ_i, all following a 1PL model with the same ability and difficulty parameters but with discrimination halved after each expansion as $\{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8} \ldots\}$. Under the SRT model, most of the statistical information from the score S_{pi} can be obtained only by a limited number of expansions. We recommend using three indicator variables since at a point where $\theta_p = \delta_i$ the three dyadic expansion fail to cover only $\frac{1}{64}$-th of the information in the continuous response.[3]

RT-Enabled Urnings Algorithm

Updating equations. Now that all the information is discretized, we can turn to derive updating equations for the RT-enabled Urnings algorithm. Deonovic et al. (2020) created an updating scheme which produced a limiting distribution which is unattainable analytically, thus, they used a Metropolis correction to restore the detailed balance. Here we provide a simplification to get rid of the Metropolis correction, based on the mathematical tricks introduced by Bolsinova et al. (2022).

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Assume that we created k conditionally independent indicator variables $\{X_{pi}, X_{pi}^{(2)}, \ldots, X_{pi}^{(k)}\}$ based on the dyadic expansion of the continuous score S_{pi}, all of which can be modeled with 1PL models with discriminations $\{1, \frac{1}{2}, \ldots, \frac{1}{2^k}\}$. Since these indicator variables are conditionally independent, we can treat them as responses to separate items with different discriminations. Now, the goal is to construct an updating scheme using these variables that forms a Markov chain with a finite discrete state space but takes the discrimination of the variables into account. The issue is that the discriminations are not integers. To circumvent this, we rescale the variance of the abilities and difficulties as $\theta_p^* = \theta_p/2^k$ and $\delta_i^* = \delta_i/2^k$. This way, the discrimination parameters can be rescaled to $\{2^k, 2^{k-1}, \ldots, 1\}$.

To derive the updating equations we again start with expit-transforming the ability and the difficulty as $\pi_p = \frac{\exp(\theta_p^*)}{1+\exp(\theta_p^*)}$ and $\xi_i = \frac{\exp(\delta_i^*)}{1+\exp(\delta_i^*)}$. Using the rescaled parameters we can parameterize a 1PL model for an arbitrary indicator variable $X_{pi}^{(m)}$[4] with a corresponding discrimination $w \in \{2^k, 2^{k-1}, \ldots, 1\}$ as

$$P(X_{pi}^{(m)} = 1 \mid \pi_p, \xi_i) = \frac{\pi_p^w(1-\xi_i)^w}{\pi_p^w(1-\xi_i)^w + \xi_i^w(1-\pi_p)^w}. \tag{13}$$

The probability of observing $X_{pi}^{(m)} = 1$ is analogous to drawing w green balls from an infinitely large urn (representing the ability of the student) and w red balls from an infinitely large urn (representing the difficulty of the item), where we only accept draws if exactly w green balls were drawn from one of the urns and w red balls from the other one.

Similar to the original Urnings algorithm, we will track ratings defined on a discrete grid of proportions (analogous to using finite-sized tracking urns), but the updating needs to account for the difference in discrimination of the indicator variables. The idea is that the response of an item with a larger discrimination parameter can affect a larger change in the ratings. We define the updating scheme such as the student ratings can change by $\frac{w}{n_p}$, 0 or $-\frac{w}{n_p}$ with a corresponding symmetric item update $-\frac{w}{n_p}$, 0 or $\frac{w}{n_p}$:

$$\begin{aligned} \frac{R_{p,t}}{n_p} &= \frac{R_{p,t-1}}{n_p} + \frac{w}{n_p}\left(X_{pi}^{(m)} - X_{pi}^{*(m)}\right), \\ \frac{U_{i,t}}{n_i} &= \frac{U_{i,t-1}}{n_i} + \frac{w}{n_i}\left(X_{pi}^{*(m)} - X_{pi}^{(m)}\right), \end{aligned} \tag{14}$$

where $X_{pi}^{*(m)}$ is the simulated response. To obtain the simulated response, we first add w green balls to the student's tracking urn and w red balls to the item's tracking urn if the response is correct, and vice versa if the response is incorrect, which can be

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represented as

$$\begin{aligned} R_p^* &= R_{p,t-1} + wX_{pi}^{(m)}, \\ U_i^* &= U_{i,t-1} + w(1 - X_{pi}^{(m)}), \\ n_p^* &= n_p + w, \\ n_i^* &= ni + w. \end{aligned} \tag{15}$$

Then the probability of observing $X_{pi}^{*(m)} = 1$ is analogous with drawing exactly w green balls from the students' augmented tracking urn and w red balls from the items's augmented tracking urn without replacement, where the only accepted draws are when exactly w green balls were drawn from one of the urns and w red balls from the other one. The probability of drawing a given amount of green balls from an urn without replacement can be modeled with Hypergeometric distributions. Thus, the probability of a correct simulated response[5] based on the ratings can be characterized as

$$\begin{aligned} Z_p &\sim Hypergeometric(w, n_p^*, R_p^*) \\ Z_i &\sim Hypergeometric(w, n_i^*, U_i^*) \\ P(X_{pi}^{*(m)} = 1 \mid R_p^*, U_i^*, n_p, n_i) &= \frac{p(Z_p = w)p(Z_i = 0)}{p(Z_p = w)p(Z_i = 0) + p(Z_p = 0)p(Z_i = w)} \end{aligned} \tag{16}$$

$$= \frac{\prod_{v=0}^{w-1}(R_p^* - v)\prod_{v=1}^{w}(n_i - U_i^* + v)}{\prod_{v=0}^{w-1}(R_p^* - v)\prod_{v=1}^{w}(n_i - U_i^* + v) + \prod_{v=0}^{w-1}(U_i^* - v)\prod_{v=1}^{w}(n_p - R_p^* + v)}.$$

The resulting updating scheme is a special case of the updating scheme of the Multidimensional Urnings algorithm (Bolsinova et al., 2022). The resulting Markov chain has a limiting distribution which is a product of non-identical binomial distributions conditional on the total sum of the ratings in the system[6].

Two Possible Algorithms. Deonovic et al. (2020) used multiple ratings to take all indicator variables into account. We will expand on this idea because their formulation does not allow for adaptive item selection, and it is unclear how the multiple ratings can be combined to get a single rating containing information from both accuracy and response times. Apart from their discussion of the algorithm lacking the necessary detail to reproduce the algorithm, they also did not account for the fact that updating a rating based on an indicator variable with a lower discrimination value will converge to its limiting distribution more slowly, compared to one with a higher discrimination value. This is problematic because combining the separate ratings into a single one would be biased if the rating based on the higher-order indicator variables has not converged.

First, we present our modification of their algorithm, and then we provide an alternative way to incorporate the indicator variables. We will refer to the former as the Multiple Urns algorithm (Figure 2) and the latter as the Multiple Updates algorithm (Figure 3).

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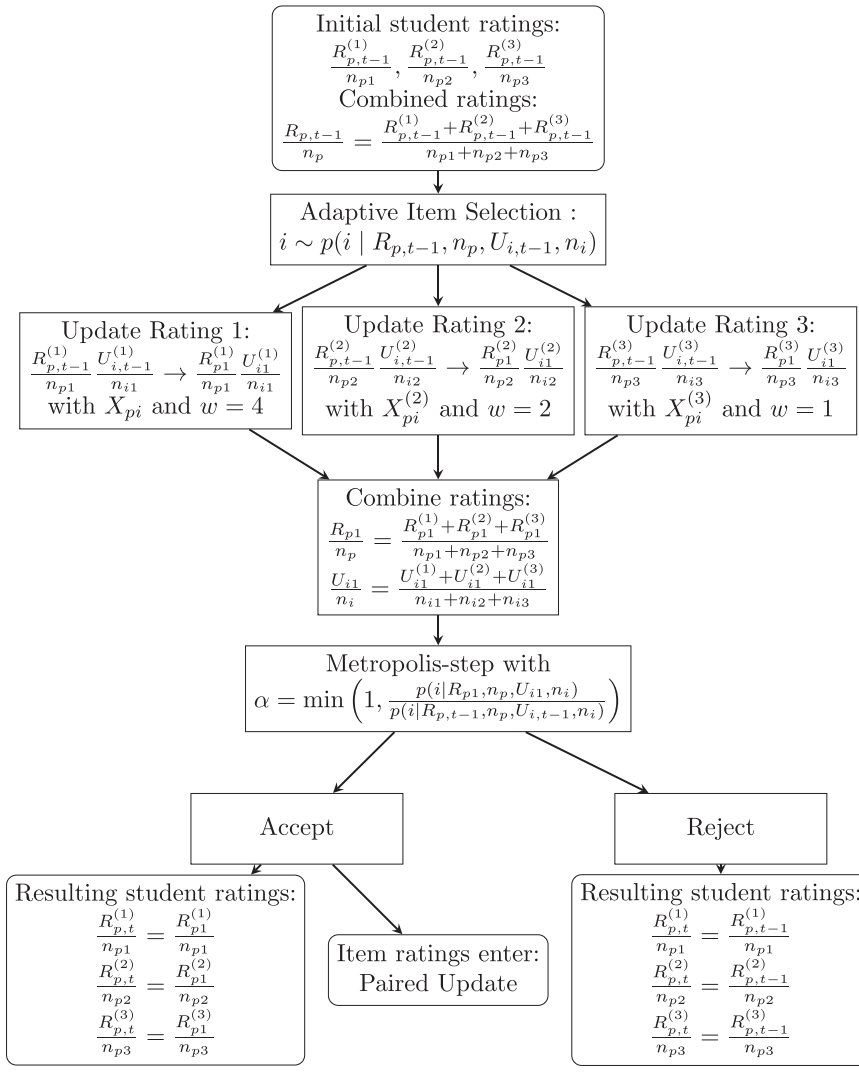


Figure 2. Flowchart of the Multiple Urns algorithm including adaptive item selection, the Metropolis-step and the item paired updates.

In the Multiple Urns case, we first decide how many indicator variables will be defined, and we assign a rating for each of the variables. Assuming a three-order dyadic expansion, the first ratings track the ability/difficulty based on the accuracy, and the second and third ratings track the ability/difficulty based on the first pseudo-RT and second pseudo-RT. Since the update sizes differ among the ratings, the indicator variables with smaller discriminations move through the parameter space more slowly, so they need a lot more data to converge. Gergely et al. (2024) obtained that doubling the urn size requires approximately twice the amount of item response to converge, so we recommend to set the urn sizes for the tracking urns as $\{n_{p1}, n_{p2} = n_{p1}/2, n_{p1}/4 \ldots, n_{p1}/2^k\}$ in practical application to ensure comparable

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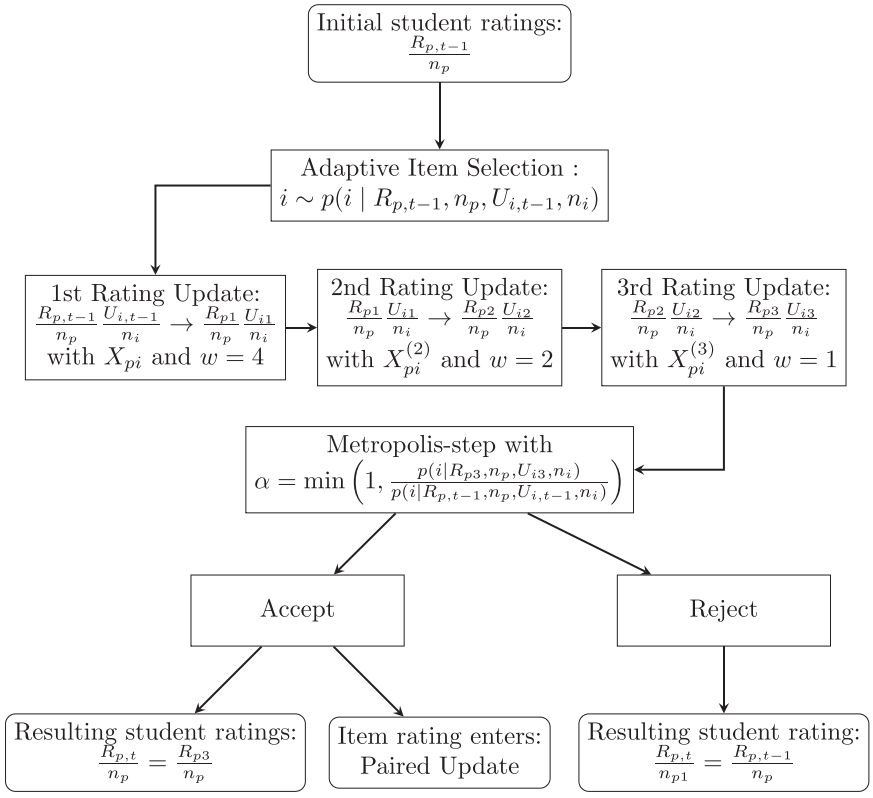


Figure 3. Flowchart of the Multiple Updates algorithm including adaptive item selection, the Metropolis-step and the item paired updates.

convergence rates. After the ratings of all the tracking urns converge to their limiting distribution, we can sum the numerator ratings and divide with the sum of urn sizes to obtain a single estimate for π_p[7]. Since all the ratings are conditionally independent and converge to a binomial distribution with a probability of π_p and size equal to the corresponding urn size, their sum is also a binomial distribution with size equal to the sum of the urn sizes.

There are multiple options to adaptively select an item. One could stipulate the item selection probability distribution solely on the rating based on accuracy to match the adaptive item selection of the original Urnings algorithm. However, in this case, any potential improvement of measurement precision obtained from response time will not be utilized for improving the quality of item selection. Therefore, here we instead first combine the different ratings into a single rating so that a higher precision ability measure is obtained, allowing a more optimal item selection.

After the item selection, the three indicator variables are calculated from the observed accuracy/RT and each of the ratings is updated based on its corresponding indicator variable and its discrimination. Since adaptive item selection destroys the detailed balance of the updating scheme, we need to use a Metropolis step with

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acceptance probability calculated similarly to Equation 4. Since the adaptive item selection was based on the combined ratings, we compare the combination of the proposed updates compared to the combination of the previous ratings. If the proposal is rejected, then all updates for all three ratings are undone.

Since combining the ratings in the Multiple Urn algorithm when the ratings are not yet converged may result in biased estimates we propose a way to incorporate the information from all indicator variables into a single rating. In the Multiple Updates case, we define a single tracking urn for each student. Similar to the Multiple Urns case we first adaptively select an item, and calculate the indicator variables. The rating is first updated based on accuracy with a discrimination of 4. Then the new rating is updated based on the first pseudo-RT with a discrimination of 2, and the resulting rating is updated based on the second pseudo-RT with a discrimination of 1. So in total the rating can change between $\frac{7}{n_p}$ and $-\frac{7}{n_p}$. Lastly, the rating is accepted/rejected using a Metropolis step.

Paired Update Procedure. If the abilities on average grow in the system, that would mean that the item difficulties, due to the symmetric nature of the updates, necessarily decrease on average. That would make the comparison between time points impossible since the scale on which we measure the students is constantly changing (Brinkhuis et al., 2015; Bolsinova et al., 2024; Hofman et al., 2020). To fix the latent scale, we introduce a generalized version of the paired update procedure (Bolsinova et al., 2022). The idea behind paired update is to keep the total sum of green balls in the item bank constant so that the change on the student side can be interpreted as a change with respect to the item bank. To do this, items are not immediately updated, but they first enter a queue. When the Multiple Urns algorithm is used, for each item rating, we proceed with the following algorithm. If an item obtains a negative update based on a given indicator variable, it only gets updated if there is at least one item waiting for a positive update based on the same indicator variable. Otherwise, it enters the queue and is not updated until a positive update comes in. However, when the Multiple Updates algorithm is used, the updates are not of the same size, so the paired update needs to account for the size of the update too.

This modified paired update procedure operates as follows. If we obtain an update of size $\frac{w}{n_i}$ for item i we search for candidate items with negative updates that sum to $-\frac{w}{n_i}$. To obtain an update of size $-\frac{w}{n_i}$, it is possible to choose multiple candidate items if the sum of their queued updates is equal to $-\frac{w}{n_i}$. If there are item combinations that satisfy this condition, then item i and all the candidate items are updated accordingly. If there are no suitable candidate items, item i enters the queue with $\frac{w}{n_i}$ update size[8].

Simulation Study

We used simulation studies to compare the performance of the RT-enabled Urnings algorithm to the original Urnings algorithm. We hypothesized that when the abilities are stable, the RT-enabled Urnings algorithm requires fewer responses to reach the same level of measurement error, and given a predetermined number of responses, it can attain a lower error level. Similarly, when the abilities are changing, the RT-enabled Urnings algorithm can track the changes more accurately given the same urn size.

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The routines for running a single system with different versions of the Urnings algorithm were written in C (Ritchie et al., 1978), while the simulations were performed in R (R Core Team, 2023). For data wrangling and plotting of the results, we used the Tidyverse package (Wickham et al., 2019).

The codebase and all the resulting simulated data are available on GitHub and OSF.[9]

Simulation with Stable Abilities

First, we simulated systems where the abilities were stable, but we did not have prior information about the abilities of the students and the item difficulties were not estimated before the start of the system, meaning that we initialized the systems from a cold start (see section *Cold Start*). Second, we tested how individual-level ratings converge when the system is calibrated and only a few new students enter the system (see section *Calibrated System Simulation*).

In these simulations, we created systems with 1,000 students and 200 items, and each system was replicated multiple times. In the cold start case, each system was replicated 1,000 times, while in the calibrated case, we replicated each system 10,000 times so that we can calculate reliable outcome measures at the individual level. In the cold start case, each student responded 500 times to adaptively selected items, while in the calibrated case, this number was reduced to 250 to increase the runtime and decrease the memory pressure of the simulation. For adaptive item selection, we used the Normal Kernel Method (Bolsinova et al., 2022; Gergely et al., 2024). The probability of selecting item i for student p is proportional to the difference between their logit-transformed ratings as:

$$P(i \mid R_p, n_p, R_i, n_i) \propto \exp\left(-\frac{1}{2\sigma_p^2}\left[\left(\ln\left(\frac{R_{p,t-1}+1}{n_p - R_{p,t-1}+1}\right) - \ln\left(\frac{R_{i,t-1}+1}{n_i - R_{i,t-1}+1}\right)\right) - \mu_p\right]^2\right), \tag{17}$$

where μ_p is the preferred probability of correct solution and σ_p is the dispersion parameter, which allows some variability of the selection probabilities. To obtain a probability distribution, this kernel is normalized with the sum of the selection probabilities of the item bank. The adaptive item selection was parameterized such that items are more likely to be selected if the student has a $\mu_p = 0.7$ probability of correctly solving them, given their current rating, and the dispersion of the selection probability was $\sigma_p = 0.5$. The mean of the abilities were $\mu_\theta = \ln(0.7/0.3)$, with a standard deviation of $\sigma_\theta = 1$, whereas the mean of the item difficulties were set to $\mu_\delta = 0$ with a standard deviation of $\sigma_\theta = 2$. Then the true abilities/difficulties were set as equal quantiles of these normal distributions. This way, an average student had a 0.7 probability of solving an average item, which fits the preferred probability of correct response of the adaptive item selection.

We used a third-order dyadic expansion for the RT-enabled Urnings algorithms, resulting in 3 indicator variables: the accuracy, the first pseudo-RT and the second pseudo-RT. The corresponding discrimination parameters were {4, 2, 1} respectively, therefore, the population variance of the abilities and difficulties was divided by 4.

For all variants of the Urnings algorithm, the mean of the item ratings was fixed using paired updates. To make the results of the original and RT-enabled Urnings algorithms comparable, we used the same urn sizes. The item urn sizes were fixed

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throughout the simulations to 350. We tested 3 different student urn sizes: 70, 140, and 280. In the case of the original Urnings algorithm, it means that if we use an urn size of 350, instead of updating the abilities/difficulties with $\frac{1}{350}$ after each response, we updated with $\frac{4}{350}$ to match the discrimination of the weight of the accuracy in the RT-enabled cases. When the Multiple Urn variant was used, we separated the urn size used in the other cases into 3 parts in a 4 to 2 to 1 ratio (i.e., an urn size of 350 is separated as 200 for the accuracy, 100 for the first-order pseudo-RT, and 50 for the second-order pseudo-RT). This was done to accommodate for dependence of convergence speed on urn size. The exact ratio was selected based on the results of (Gergely et al., 2024).

In the first simulation (cold-start case), the initial rating for both the students and the items was set to 0.5. In the second simulation, we acted as if students were practicing for a longer period already, and the items were previously calibrated, meaning that their initial rating was sampled from their corresponding limiting distribution, which is a binomial given their true ability/difficulty and the urn size. In this case, we added 8 new students to the system with true abilities equally divided from –2 to 2 on the logit scale, and without prior information about their ability, thus a 0.5 initial rating.

We used three outcome measures: the mean squared error (MSE) between the student ratings and the true values in every time point, the system-level hitting time (HT_{system}) and the student-level hitting time $HT_{student}$.

The MSE was calculated using the ratings and the rescaled and expit-transformed true abilities as:

$$\mathrm{MSE}_t = \sum_{p=1}^{\#students} \left(\frac{R_{p,t}}{n_p} - \frac{\exp(\theta_p^*)}{1+\exp(\theta_p^*)} \right)^2, \tag{18}$$

where $\#students$ is the number of students in the system and θ_p^* is the rescaled true ability of student p.

The goal of a system-level hitting time is to see how fast the whole system converges to its equilibrium from a cold start. To calculate the HT_{system} measure, first, we needed to set a baseline MSE. The baseline MSE refers to the MSE of a system which reached equilibrium, with the same abilities/ difficulties and urn sizes as the simulated systems. To achieve this, we first simulated ratings for all the students based on their corresponding limiting distribution (binomial with the urn size and the expit-transformed ability), and we calculated the MSE. This process was replicated 10,000 times, thus a distribution of MSE was obtained. Here we calculated the 2.5% and 97.5% quantiles of this distribution, which will be referred to as the 95% central interval of the sampling distribution. These intervals were calculated for all three urn sizes.

The hitting time of a simulated system is the first iteration when the MSE of the simulated system is contained in the 95% central interval of the sampling distribution of a converged system. Since each simulated system in the cold start case was replicated 1,000 times, 1,000 hitting time values were obtained. We present the average of these hitting times, with their corresponding 95% central interval.

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The goal of calculating the student-level hitting times is to see how fast the convergence is at the individual level. Since ratings of a single system are volatile, we were interested in how fast the mean of each new student's ratings (across replications) reached the 95% central interval of their limiting distribution. To achieve this, we sampled 10,000 ratings for each new student and each student urn size from their corresponding limiting distribution. Then the 2.5% and 97.5% quantiles were calculated, creating a 95% central interval for the mean of the limiting distribution of all new students and urn sizes separately.

Simulation with Changing Abilities

In the third simulation, we assumed that the abilities of the students are changing linearly on the logit scale. We restricted the analysis to linear change since (Gergely et al., 2024) showed that the Urnings algorithm tracks discrete changes similarly to linear ones. Similar to the cold start simulation, we performed 1,000 replications of the systems, but we increased the number of urn sizes to see which urn size provides the best tracking capability.

We simulated systems with 1,500 students and 200 items, where each student solved 500 adaptively selected items. The adaptive item selection procedure was identical to the first two simulations.

The difficulty of the items was selected as equal quantiles of a normal distribution with $\mu_\delta = 0$ and $\sigma_\delta = 2$. The items were treated as they were calibrated before, meaning that all initial ratings for the items were simulated from their respective limiting distribution. This way, we decoupled the effect of cold start on the convergence speed, focusing solely on whether a rating with a given urn size starts to lag behind the true ability change throughout 500 items. The total changes were set to $\{-0.5, 0, 0.5, 1, 1.5, 2\}$ logits. This way, the average change in the system was overall positive, which is expected from an ALS, but we also modeled students with decreasing abilities, which is analogous to forgetting. The size of the changes was selected based on the changes observed in the MathGarden system over 1 year (Brinkhuis et al., 2018). These changes were equally distributed across students with respect to their abilities. We first set the ability distribution of the students after solving 250 items as equal quantiles of a normal distribution with $\mu_\theta = \ln(0.7/0.3)$ and $\sigma_\theta = 1$. The ability in the rest of the iterations was calculated by adding/subtracting the total change divided by the number of iterations after each iteration. The ratings of the students were initially sampled from their limiting distribution with their corresponding ability in the first iteration and the given urn size, to eliminate the effect of cold start on the tracking capability.

We used the same item urn size as before. Regarding the student urn sizes, we included all urn sizes between 70 and 560, divisible by 35, resulting in 15 urn sizes.

We used two types of outcome measures. MSE was calculated between the ratings and the expit-transformed and rescaled true values in each iteration. Based on the MSE, we evaluated each algorithm and urn sizes averaged across the change amounts. To assess whether the urn size recommendations would differ among different change amounts and algorithms, we calculated the mean of the MSEs in the

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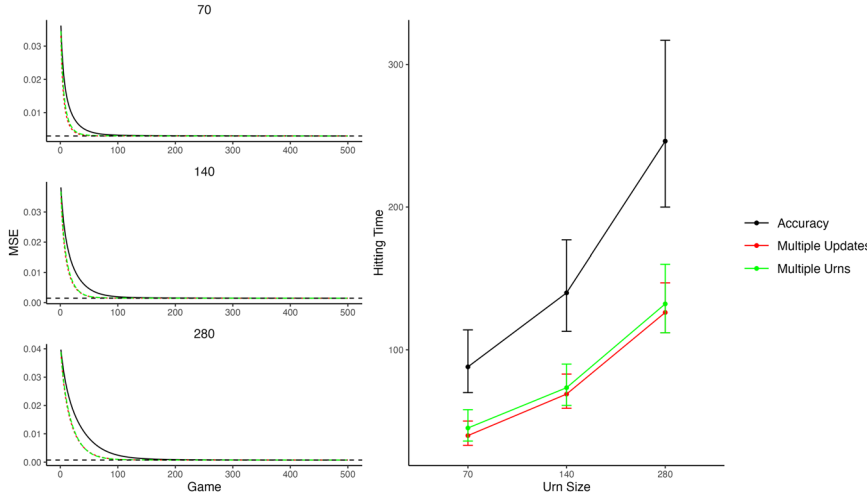


Figure 4. System-level MSE across iterations (left), and the average hitting time with the corresponding 95% central interval (right) represented as error bars of the different variants of the Urnings algorithm and student urn sizes. The dotted line represents the MSE of a converged system.

last 200 iterations and reported the urn size which resulted in the lowest mean MSE for each change amount, urn size and algorithm type.

Results

Simulations with Stable Abilities

Cold start. In the total cold start scenario, we found that the RT-enabled Urnings algorithms outperformed the original Urnings algorithm in all three student urn size cases we tested. All three algorithms converged to the baseline MSE, but the speed of convergence was different. The MSE values decreased faster with the RT-enabled variants (see Figure 4) and the average of the MSE in the first 100 iterations across the different urn sizes was 0.009 for Accuracy, and 0.006 for both the Multiple Updates and Multiple Urns, where the theoretical MSE (averaged across urn sizes) was 0.002. The Multiple Updates and Multiple Urns algorithms performed equivalently in terms of MSE.

The RT-enabled variants converged to the baseline MSE approximately twice as fast as the original Urnings algorithm. The average hitting time for the Accuracy case was 158, while the Multiple Updates algorithm converged in 78.3 and the Multiple Urns algorithm in 83.6 iterations. The Multiple Updates algorithm with a student urn size of 70 achieved the fastest convergence since the algorithm needed an average of 39.84 [33, 55.025] iterations to converge. The slowest convergence was obtained using the original Urnings algorithm with an urn size of 280, as it needed 246.233 [199.975, 317] iterations to converge. There were marginal differences in the mean HT-s between Multiple Updates and Multiple Urns variants; however, this difference is not statistically relevant since the 95% central intervals did overlap. If one wants

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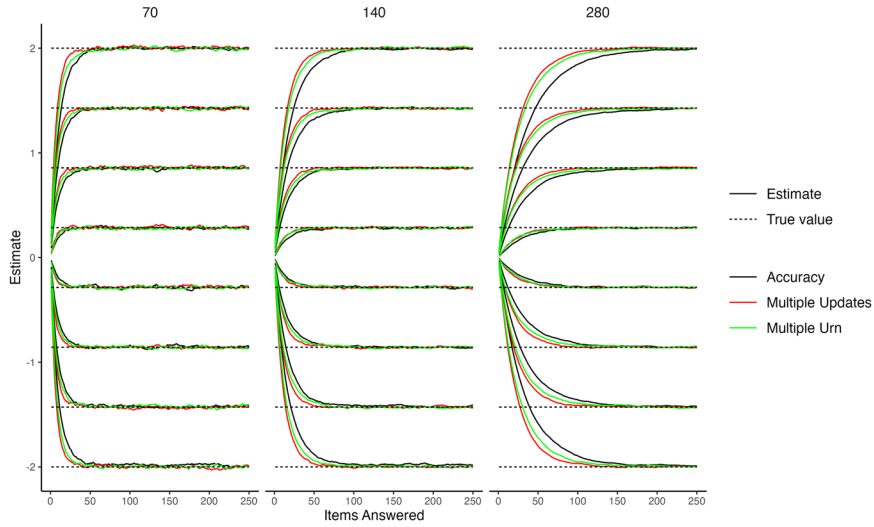


Figure 5. Average logit transformed and rescaled rating of 8 new students throughout solving 500 items using different Urnings algorithm variants.

to achieve convergence after a certain number of iterations, one can use a bigger urn size (and hence have higher precision) with RT-enabled algorithms than the original algorithm. The MSE after 50 iterations with an urn size of 70 using the original algorithm was 0.004. This error level can be reached using an urn size of 280 with the Multiple Updates algorithm.

Calibrated system. When we used a calibrated system, we found that regardless of the true ability of the student or the urn size, both RT-enabled algorithms reached the true ability faster than the original Urnings algorithm (see Figure 5). Students needed to solve an average (over urn sizes and true abilities) of 44.3, 53.8, and 68.6 items when the Multiple Updates, Multiple Urn and original Urnings algorithms were used, respectively. All algorithms provided an approximately twice as large HT when the urn size was doubled. The minimum of the HT averaged across true abilities was 17.6. It occurred with an urn size of 70 and the Multiple Updates algorithm, while the maximum HT was 121 using an urn size of 280 and the original Urnings algorithm.

The HT-s were consistently smaller with the RT-enabled Urnings variants compared to the original Urnings algorithm for all urn sizes and all true values (see Table 1). The RT-enabled algorithms achieved approximately 1.5 times faster convergence compared to using only accuracy. At the student level, the Multiple Updates algorithm yielded consistently better results than the Multiple Urns algorithm. This difference is more accentuated when larger urn sizes are used or larger distances need to be covered by the updates (see Figure 5). The difference between the HT-s of Multiple Updates and Multiple Urns when the true value was 2 logits is 12 iterations with an urn size of 70. This distance is 31 iterations when an urn size of 280 was used.

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Table 1
Student-Level Hitting Times Grouped by the Different Urn Sizes and the Different True Values of the New Students

Urn size	Algorithm	−2	−1.43	−0.86	−0.29	0.29	0.86	1.43	2
70	Accuracy	40	29	24	16	16	28	33	41
140	Accuracy	79	60	51	29	31	51	72	80
280	Accuracy	153	129	103	67	63	127	153	172
70	Multiple Updates	24	19	17	9	9	17	21	25
140	Multiple Updates	46	41	36	20	23	34	44	54
280	Multiple Updates	106	81	70	49	45	75	94	104
70	Multiple Urn	27	24	21	10	11	21	26	37
140	Multiple Urn	57	56	43	26	24	43	52	59
280	Multiple Urn	119	106	88	51	51	95	110	135

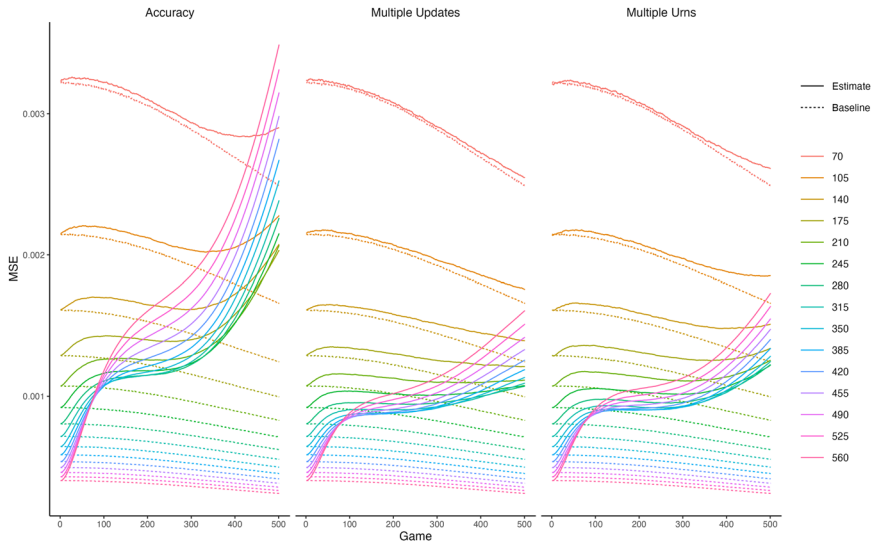


Figure 6. MSE of the different variants of the Urnings algorithm for different student urn sizes averaged across the direction and amount of changes. The dotted lines represent the MSE of a system that is in a converged state in every iteration.

Simulation with Changing Abilities

When changing abilities were used initially, the system was calibrated, which means that the initial MSE was lower for larger urn sizes. It is important to note that the baseline MSEs showed a decreasing trend over iterations. This is due to the transformation of the scale (see the dotted lines in Figure 6). The distance between abilities in a [0,1] scale is the largest at around 0.5, and as we approach 1 or 0, these distances decrease. Similarly, the largest variance of the ratings occurs around 0.5, and since the estimates are unbiased, the MSE is the mean of the rating variances.

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Table 2
Urn Sizes That Provided the Lowest Average MSE over the Last 200 Iterations for Each Algorithm Type and Amount of Change

Amount of Change	Accuracy	Multiple Stakes	Multiple Urns
−0.5	280	490	420
0.0	490	560	560
0.5	560	560	560
1.0	315	350	350
1.5	175	245	245
2.0	105	175	175

Since the abilities are on average growing, this results in a baseline MSE which is decreasing.

We obtained that using the original Urnings algorithm, after approximately 100 iterations, the MSE of systems with urn sizes between 245 and 560 started to increase substantially, showing that here the ratings are increasingly lagging behind the true ability change (see Figure 6). Throughout 500 iterations, all urn sizes showed at least some increasing trend, which means that over time, all ratings, no matter what urn sizes we used, started to lag behind the true abilities; neither urn size provided estimates which are contained in the 95% central intervals of the baseline in all iterations.

With Multiple Updates and Multiple Urns, we found similar increasing trends after 100 iterations; however, fewer urn sizes were as affected by it. In the Multiple Updates case, urn sizes up to 175 showed a marginal increase in MSE during the last 100 iterations. In the Multiple Urns case, the last urn size that only had a marginal increase was 105. In the RT-enabled cases, the estimates using the urn sizes 70 and 105 remained in the 95% central interval of the baseline MSE in all iterations. Here we also found a difference between the MSE of Multiple Updates and Multiple Urn, especially for the later iterations.

To gain insight into how different amounts of change affect the choice of the optimal urn sizes, we calculated the average MSE of the last 200 iterations for all amounts of change, all algorithm variants and urn sizes. In Table 2, we show the urn sizes for each algorithm and the amount of change which obtained the smallest average MSE. The best-performing urn sizes for Multiple Updates and Multiple Urns were consistently larger compared to the original algorithm. On average, the RT-enabled 1.37 times larger urn sizes reached the lowest MSE result. In the cases where the change was 0.5 logits or no change occurred, the best-performing urns were the maximum of the simulated urn sizes for all three algorithms, that is, with the range of urn sizes that we used, we cannot see a difference between the algorithms.

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Empirical Example

So far, we have simulated systems that use the RT-enabled Urnings algorithms and the SRT scoring rule as their measurement model and as the basis of adaptive item selection. In this empirical example, we ought to present a case when no explicit scoring rule was used within the ALS, to show that the outlined discretization strategy can still be utilized to improve measurement of ability if an appropriate way of defining fast and slow responses can be found.

We used data from an ALS called *practiceanatomy.com*. The learning system was built for practicing the recognition and correct recall of the names of anatomical structures. The system is available in English and Czech, and the majority of the users are from the Czech Republic (89%). The dataset used adaptive item selection algorithms described in Papousek et al. (2014) and Papoušek and Pelánek (2015). The data set captures the users' responses from 16 November 2015 to 3 January 2017. The total data consisted of 18,563 students, 2,894 items and 1,182,065 responses. However, we filtered students with fewer than 10 responses and items with fewer than 100 responses. The reason is that 10 responses on the student side are not enough to obtain converged ratings, even if the ability is stable, and we start the algorithm from a cold start scenario, as there is no information available about the item difficulties in this system. To reduce the heterogeneity in the sample of students and in the item pool, we only included items answered by students from the Czech Republic, and we filtered for the items where students were asked to find the given anatomical structure on the image.

The resulting dataset had 6,186 students, 1,664 items, and 459,472 responses. The proportion of correct responses was 0.75, which is similar to the proportion correct we obtained in the simulation studies using the adaptive item selection with $\mu_p = 0.7$. The median of the number of items answered by students was 29 ($IQR = 51$), and the median of the number of students answering items was 220 ($IQR = 196$). Both response distributions were positively skewed.

In addition to running the original Urnings algorithm, we ran the RT-enabled Urnings algorithms on the data, defining only first-order pseudo-RT variables, using three different ways of defining fast and slow responses. The reason was that if one can define even a first-order pseudo-RT variable which fits a 1PL model, it is still possible to augment the Urnings algorithm with response times even if they were not originally part of the scoring rule. We hypothesized that including a single pseudo-RT variable would result in better measurement of the ability, which we evaluated through the accuracy of predicting the next response (correct vs. incorrect).

Since in this data the items did not have a time limit, we used different methods to define fast and slow responses. Despite defining the cutpoint based on the median being rather arbitrary, this method has often been used in similar situations (De Boeck & Partchev, 2012; Partchev & De Boeck, 2012). We defined two median-based cutpoints as the median of all RTs (grand median) and the median RT of the given item (item median). In the third case, we tried to approximate what an appropriate item time limit would be based on the RT distribution. We defined the time limit as the response time most people needed to respond to an item, but instead of using the maximum RT, we used the 90th quantile of a given item's RT distribution, since it is

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more robust to outliers (often caused by students querying the next item, but leaving the item unanswered for a long time). Now, each item having a time limit we took half of the range between 0 and the limit (90th quantile). The cutpoints based on the median resulted in 0.5 proportion of fast responses by definition, while the 90th quantile-based method resulted in 0.66 proportion of fast responses.

All three Urnings algorithms used 60 as the item urn size (40 for accuracy information and 20 for the pseudo-RT in the case of the Multiple Urn algorithm). The student urn sizes for accuracy were chosen as a sequence of numbers divisible by 4 from 20 to 160. In the Multiple Urn cases, we added a second urn with half the urn size, while in the Multiple update case, we multiplied the urn size by 1.5. The reason for matching the size of the urns representing the accuracy is to see whether including RT in the measurement can improve the prediction of accuracy.

We evaluated the different algorithms and urn sizes based on their prediction error for the next response. The prediction error was calculated by first simulating a response and then substituting the ratings of a given student and item into Equation 1 without the urn augmentation of the original Urnings algorithm's simulated response. After each timepoint, we calculated the squared error between observed and predicted responses based on all three algorithms. Based on the recommendation of Pelánek (2015), we calculated the student-level mean squared error, which means that we first averaged the prediction error for all students separately, and then we averaged to student averages to obtain the final metric. This way we reduced the effect of students solving an extreme amount of items, and thus we created a more conservative, but student-focused metric. Note that using the Multiple Urn version of the RT-enabled Urnings, we first combined the urns to get a single ability/difficulty estimate. We replicated each analysis 100 times to reduce noise in the prediction errors, reporting the average prediction error over the replications.

We found that the Multiple Updates algorithm produced the smallest prediction error in all three cases, and it was consistently better than using only accuracy or the Multiple Urns algorithm (see Figure 7). Contrary to this, the Multiple Urns algorithm almost always yielded worse prediction errors with the grand median and item median cutpoint compared to both accuracy and the Multiple Updates algorithm. Overall, the smallest prediction error (0.192) was reached using the 90th quantile rule with the Multiple Updates algorithm and an urn size of 60 (which is a sum of 40 for accuracy + 20 for pseudo-RT). Compared to that, using accuracy and the Multiple Urn algorithm, the smallest prediction error was 0.197 and 0.194 with urn sizes 44 and 54 (36 for accuracy and 18 for pseudo-RT), respectively. Therefore, it is possible to reach a smaller prediction error when we include RT. In the case of the median-based cutpoints, the decrease in prediction error was smaller when Multiple Updates were used compared to the 90th quartile method. This shows that the median-based definition of fast/slow responses is improving the prediction accuracy to a lesser extent than the 90th quantile-based method, compared to using solely the accuracy.

Next, we evaluate how well the model with the best prediction error fits the data. To evaluate the model fit, we used the model fit property of the Multidimensional Urnings algorithm (see Bolsinova et al., 2022, Theorem 2). They state that if the measurement model holds, then the proportion of observed scores of 1 is equivalent

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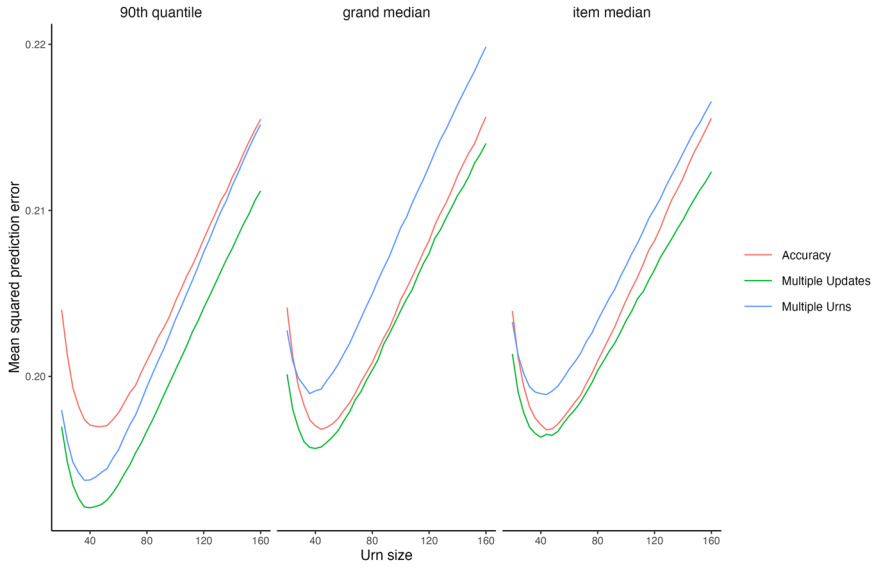


Figure 7. The mean squared prediction error of the three variants of the Urnings algorithm, with three methods to define fast/slow responses. The x-axis denotes the urn size used with accuracy. The RT-enabled algorithm used a 1.5 times larger urn size since the RT was included as additional information upon the accuracy. In the Multiple Urns case, the prediction accuracy was calculated based on the combined rating.

to the proportion of simulated scores of 1 based on the augmented ratings (see Equation 15) for any combination of values for $R_p + X_{pi}$ and $U_i + (1 - X_{pi})$.

In Figure 8, we can see that both the accuracy and the pseudo-RT variable reasonably fit a 1PL model. However, in the case of accuracy, the proportion of observed correct responses is systematically higher than the expected proportions. This might be due to the ratings lagging behind the ability development. For low/high observed proportions, the expected proportions spread more compared to the observed proportions around 0.5. The fit in the pseudo-RT case is worse compared to the accuracy. For higher values, the observed proportions are higher than the expected ones and overall, with lower observed ratings, we obtained more noise in the expected ratings.

Discussion

In this paper, we presented two rating systems capable of incorporating response time into their measurement model. Instead of using the data-driven hierarchical framework for accuracy and response time, we built them on an explicit scoring rule. The advantage of using explicit scoring rules compared to the hierarchical models is that they combine the accuracy and RT into a single variable, which is easily applicable to the updating equations of rating systems. Rating systems were used to ensure that the resulting algorithms are computationally efficient and do not require an explicit model for the ability growth (compared to dynamic IRT models, see Wang et al., 2013). We expanded the Urnings algorithm rather than the more widely

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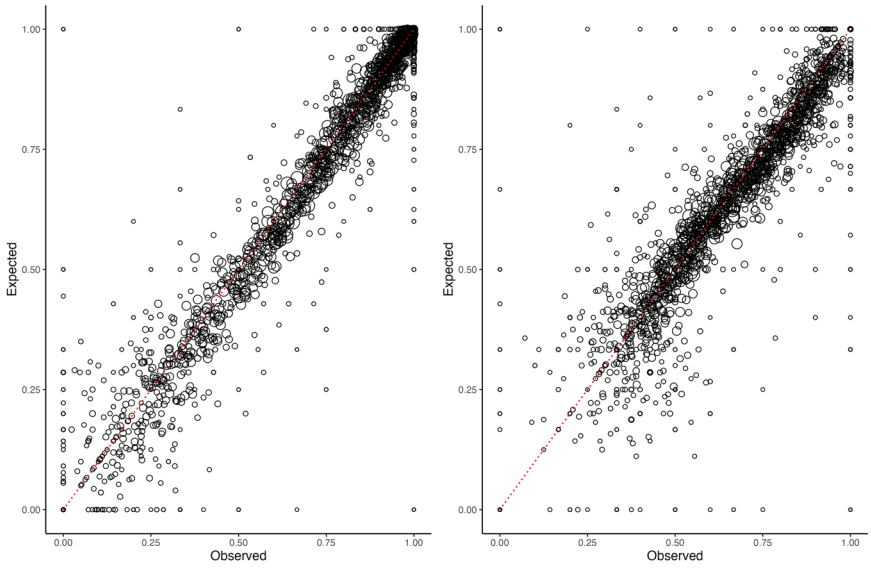


Figure 8. Evaluating the fit of 1Pl models on accuracy (left) and the first pseudo-RT variable. Each dot represents the observed (x-axis) and expected (y-axis) proportion of correct responses of a given combination of student and item ratings among all responses.

applied Elo Rating System since the Urnings algorithm provides a known limiting distribution, unbiased estimates of the ability/difficulty, and it solves the rating variance inflation when adaptive item selection is used.

Since the Urnings algorithm is inherently discrete, we relied on the discretization strategy introduced by Deonovic et al. (2020) to include the SRT scores in the updating equations; however, we significantly extended what they have proposed. First, their updating scheme required a Metropolis correction to restore the detailed balance of the Markov chain. We modified the updating scheme using the mathematical trick introduced by Bolsinova et al. (2022), thus, the resulting algorithm does not need a Metropolis step, ultimately reducing the complexity of the algorithm. Second, they discussed a simple version of the Multiple Urn algorithm, but they did not account for the difference in convergence speed between the ratings. We solved this issue by introducing different urn sizes that match the discrimination of the indicator variables that are used to update them. Third, their formulation lacks two essential features of an algorithm for ALS: adaptive item selection and an algorithm to fix the latent scale, which ensures the comparison of changing ability estimates. We provided a probabilistic way to adaptively select items for student based on their current ratings, and we corrected for the dependency caused by this selection. Additionally, we proposed an alternative approach in which a single rating is updated multiple times based on the conditionally independent indicator variables.

Using simulation studies, we tested whether including RT-s into the measurement would improve the convergence speed of the Urnings algorithm both on the system-level and the individual-level when the abilities are constant, assuming that

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the indicator variables fit their respective 1PL model. We found that the RT-enabled algorithms provide faster system-level convergence when the system is in a total cold start and a faster individual-level convergence regardless of the true abilities when new students start to practice in an ALS. These results demonstrate that adding RTs can help escape the cold start quicker and present students with better-suited items early on, which is important to keep the students engaged and to avoid the risk of drop-out. In settings where escaping the cold start quickly is not the main objective (e.g., where results from assessment can be used to set starting values), adding RTs can allow to use of larger urn sizes to achieve higher measurement precision.

The RT-enabled algorithm also provided better tracking capabilities compared to the original algorithm. We showed that the mean squared errors of the RT-enabled algorithms did not diverge as substantially from the baseline as with only using accuracy. Furthermore, we found that a higher urn size can be used to accurately track the different amounts of linear change (apart from the change of 0.5 logits, where all the algorithms performed identically). While our simulation study showcases potential benefits of RTs for the measurement of ability in ALS, it also has some limitations. First, we used a simple growth model which does not match the possible complexity in the individual learning trajectories. Second, we generated the data under the SRT model, ignoring that in practical applications, aberrant responses and model violations are possible. Extending the study to include more realistic scenarios and a thorough analysis of the effect of model misfit are both crucial next steps of this research.

It was expected that in the simulation where the true model was used to simulate observed responses, including all available information in the ability measures should at least marginally. So, to show that even when the model is not perfectly fitting and only a single pseudo-RT variable is included, the precision of the ability measures will still be higher when using RT-enabled algorithms. We presented a reanalysis of an existing ALS. Here we showed that if the original measurement model did not include the SRT scores and the items have no time limit defined, it is still possible to utilize the discretization strategy we outlined. We defined a single pseudo-RT variable indicating fast/slow responses using three heuristic methods. We found that when fast and slow responses are defined based on the median, the prediction accuracies are not substantially better compared to only using accuracy. When we mimicked a time limit by taking the 90th quantile of each item's response time distribution and defined fast and slow responses based on half of the interval between 0 and this quantile, both RT-enabled algorithms outperformed the original algorithm. This method can be used in cases when the accuracy and the pseudo-RT variables can be reasonably modeled by 1PL models. In our case, we have not found severe signs of misfit and observed that including RTs improved prediction accuracy.

Throughout the analyses, results mainly favored the Multiple Updates algorithm. This is because the combined ratings only converge if all three ratings are converged, which, on average, takes a larger number of iterations. These results might seem to indicate that there are no use cases where the Multiple Urns algorithm could be recommended; however, in many scenarios, keeping the ratings from response accuracy ("the pure measure of ability") separate from the ratings from RT-s can be advantageous, as RT-s can potentially introduce bias and construct irrelevant variance. By

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doing so, we can use an adaptive item selection policy based on accuracy and can move between different measures in different situations (the more high-stakes it is, the more careful one should be with adding RT information) (Becker et al., 2021).

When including RTs in the algorithm, its computational complexity increases compared to the simple Urnings algorithm. However, since the basic algorithm is very low-weight in terms of computations, the complexity of the extended algorithm is still low in absolute terms and in comparison to other psychometric methods and does not harm the feasibility of the approach.

Moving forward, the presented algorithms should be evaluated in more data sets, including different skills and subpopulations, as whether including RTs into the measurement of ability improves its quality depends on how well the relationship between ability and RTs postulated by the SRT model matches this relationship, which can depend on the context. It is important to consider not only the potential increase in measurement precision but also the issue of validity. Explicitly communicating the scoring rule to the respondents is one way to ensure validity. However, even then, there may be individual differences between the students in terms of how they adapt their response behaviour to the communicated scoring rule. Therefore, it is important to study whether inclusion of RTs improves measurement at least in terms of prediction accuracy of next responses, as we illustrate in our empirical example.

The SRT model uses only one latent dimension for both response accuracy and RT data. However, a single dimension might not be sufficient to explain both accuracy and RT data, and it may be beneficial to include an additional dimension which explains the differences in RTs that do not reflect the differences in ability in order to avoid possible confounding. One possible approach is to derive rating systems based on the hierarchical framework. Both the Elo rating system and the Urnings algorithm can be generalized to use multidimensional models, so it might be possible to derive rating systems tracking the ability/difficulty and the speed/time intensity latent variables. But even then, the correlation between the latent variables should be estimated before the system starts or should be updated on-the-fly. The issue of current implementations of the multidimensional Urnings algorithm is that the different dimensions are tracked independently from each other, and there is no clear mechanism to borrow information from one to another. In practice on-the-fly updating of the latent correlations would further decrease the transparency of how the ALS works. Another possibility is to develop a rating system extension of the multidimensional discretized SRT model developed by Tijmstra and Bolsinova (2021). This model uses a similar discretization strategy as we used in our study, but defines a latent speed variable, creating a bridge between scoring rules and the hierarchical modeling framework.

The SRT scoring rule defines a very specific relationship between accuracy and RT, which might not be beneficial in many circumstances. In low-stakes measurement, it might overpenalize guessing and disadvantage students who are naturally slower/more thorough at solving exercises. In these cases, other scoring rules could be applied, such as the correct item summed residual time (Van Der Maas & Wagenmakers, 2005; Vermeiren et al., 2025), which does not deduct more points for fast incorrect responses. For scoring rules other than SRT, it is unclear how the presented discretizations can be used. Furthermore, it should be evaluated whether the scoring

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rules in different subgroups fit separately, and in the case of a misfit in a specific group, a mixture model of different scoring rules is yet to be developed.

While the Urnings algorithm has been extended to multiple abilities, the discrete nature of the algorithm makes it unclear how the current RT-enabled version can be extended to more complex IRT models. Another possible direction is to formulate explicit scoring rules for other types of process data and build similar rating systems that enable the incorporation of them into the Urnings algorithm.

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Notes

[1]The detailed derivations can be found in the online Supplementary Material (https://osf.io/475kg/?view_only=26c6d2ec6d314e94b4d76057c8b85ff3). In this case, we refer the reader to the proof of Lemma 1

[2]Find the proof of Lemma 2 in the Supplementary Material.

[3]For a student-item pair where $\theta_p = \delta_i$, the ratio of information in the first three indicator is proportional to the ratio of the sum of squared discriminations $4^2 + 2^2 + 1^2 = 21$ to the total sum of $4^2 + 2^2 + 1^2 + 1/2^2 + 1/4^2 + \cdots = 64/3$, that is the unaccounted portion of information is $\frac{1}{3}/(\frac{64}{3}) = 1/64$.

[4]To unify notation we add superscript (1) to response accuracy

[5]Find the proof of Lemma 3 in the Supplementary Material.

[6]Find the proof of Theorem 1 in the Supplementary Material.

[7]This is analogous with having 3 urns with green and red balls and pouring all the content of these urns to a bigger urn.

[8]For example, if an item needs an update of $\frac{4}{n_i}$, then the paired update looks at whether there is an item in the queue with an update of $\frac{-4}{n_i}$, or a pair of items with updates $(-\frac{3}{n_i}, -\frac{1}{n_i})$ or $(-\frac{2}{n_i}, -\frac{2}{n_i})$, or 4 items with updates all equal to –1.

[9]The peer-review-friendly GitHub repository can be found here: https://anonymous.4open.science/r/RT_urnings-84C2/

All data used to produce the results can be found at this link: https://osf.io/pq3ab/?view_only=e768e7022ead4e958fb8277f3ba52ac6

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